



Hotel Reservation and Guest Service Management System

DATABASE MANAGEMENT SYSTEMS – FINAL PRESENTATION

Name: Kavuri Rohan Chowdary

Roll No: 25WU0102122

Section: CSE AI&ML

PANTHERS

Faculty: Kirin Mayee Adavala

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PROJECT OVERVIEW

A DBMS-based application for managing hotel reservations and guest services, built on MySQL with a user-friendly web interface.

Room Management &
Guest Management

Reservation
Management & Stay
Management

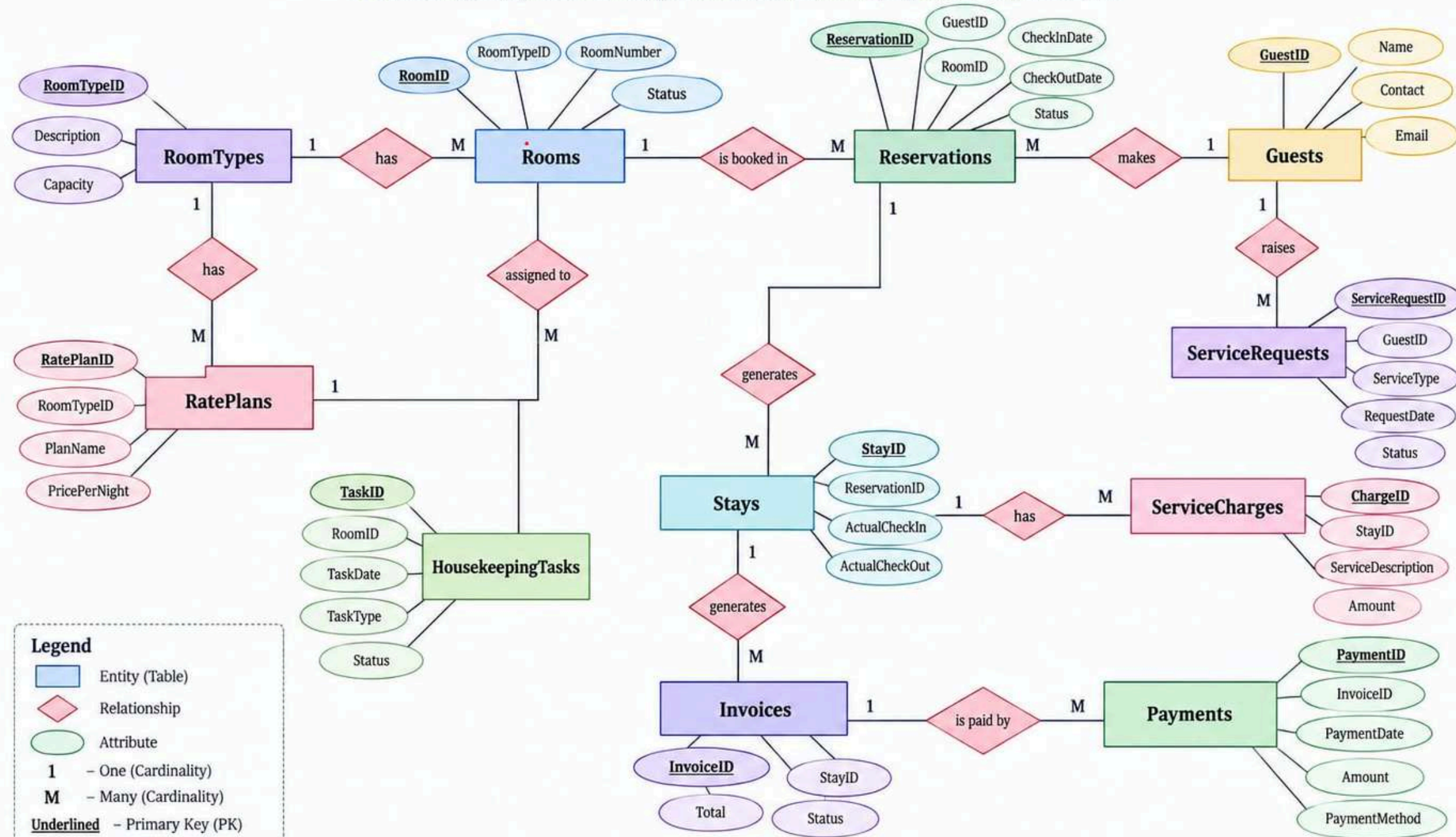
Guest Services &
Housekeeping

Billing, Payments &
Full CRUD Operations

Entity-Relationship (ER) Diagram

Hotel Reservation + Guest Services DBMS

11 entities covering reservations, guest services, housekeeping and billing workflows





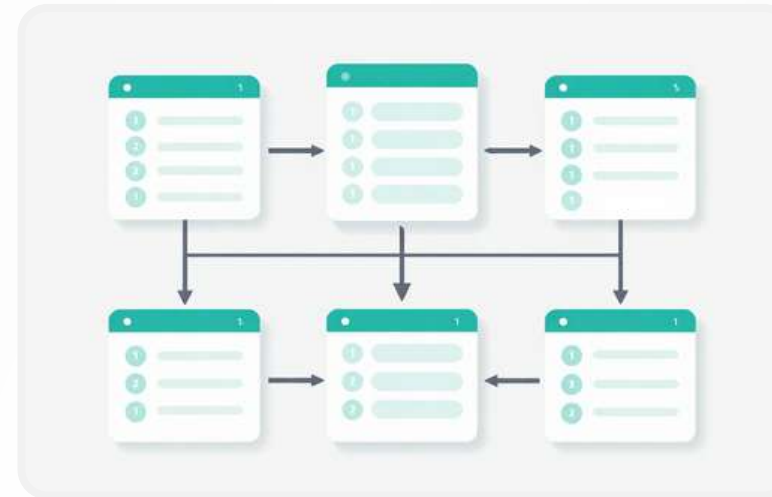
Manual hotel record management leads to errors, inconsistency, and inefficiency.

Challenges & Project Objectives

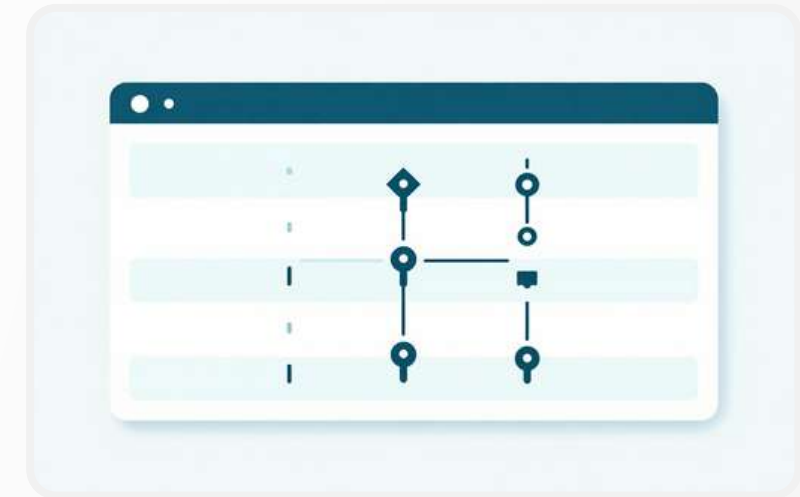
- 1 Duplicate records, data inconsistency, and errors in room availability tracking
- 2 Difficulty tracking reservations, guest stays, service requests, and payments
- 3 Centralize hotel data; manage guests, rooms, reservations, invoices via CRUD operations
- 4 Ensure data integrity through relationships, constraints, and a normalized relational structure

DATABASE DESIGN

11 relational tables with
primary keys, foreign
keys, constraints, and
normalized structure.



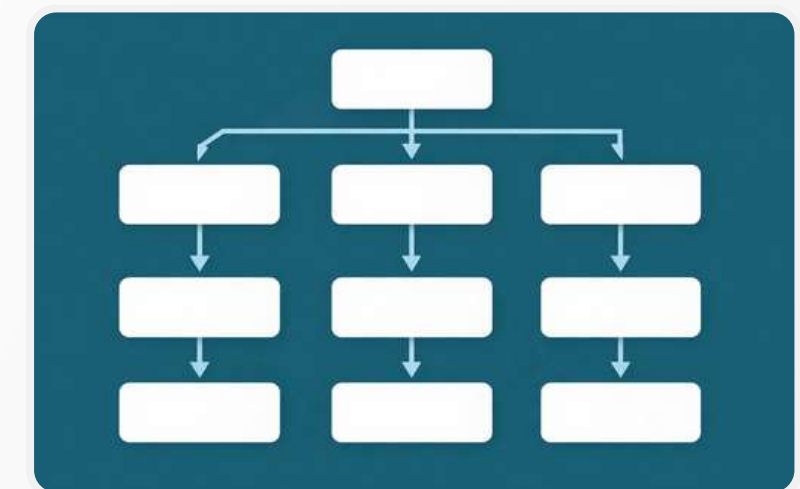
Core Tables: RoomTypes, Rooms,
RatePlans, Guests, Reservations, Stays



Extended Tables: ServiceRequests,
HousekeepingTasks, ServiceCharges,
Invoices, Payments



Constraints: Primary Keys, Foreign Keys,
NOT NULL, UNIQUE, Referential Integrity

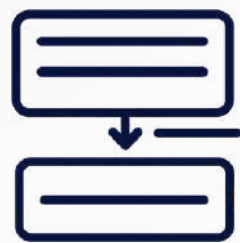


Structure: Normalized relational design —
ER Diagram placeholder

RELATIONSHIPS

Entity Relationships & Data Flow

Key relationships linking rooms, guests, reservations, stays, services, and payments across 11 relational tables.



Room Relationships
RoomTypes → Rooms
→ Reservations →
HousekeepingTasks



Guest & Reservation
Links
Guests → Reservations
← RatePlans;
Reservations → Stays



Service & Charge Flow
Guests/Stays →
ServiceRequests →
ServiceCharges



Billing Lifecycle
Stays → Invoices →
Payments

Technology Stack & Architecture

Frontend

HTML, CSS, and JavaScript power the user-facing web interface for hotel record management.

Backend & Platform

Node.js and Express.js handle server-side logic and API routing. Developed and hosted on Replit.

Database & Architecture

MySQL stores all hotel data. Flow: User → Web Interface → Frontend → Backend → MySQL → Hotel Management Tables.

USER INTERFACE

The dashboard provides a centralized web interface to access and manage all hotel modules, with real-time statistics for operational oversight.

Module Access

Room Types, Rooms, Rate Plans,
Guests, Reservations, Stays

Services & Finance

Service Requests, Housekeeping,
Service Charges, Invoices,
Payments

Guest & Room Stats

Total Guests · Total Rooms ·
Available Rooms · Total
Reservations

Operational Stats

Active Stays · Pending Services ·
Pending Housekeeping Tasks ·
Total Payments

INSERTION & VIEWING OF RECORDS

Adding and retrieving guest records through forms and SQL queries.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Open the Add Guest form with fields: First Name, Last Name, Email, Phone, Address, ID Proof	Fill in guest details and submit the form to trigger the insert operation	Record is inserted into MySQL Guests table via backend (Node.js + Express.js)	Retrieve records using: <code>SELECT * FROM Guests;</code> query executed on the database	Results displayed in tabular format showing all guest records before and after insertion

Deletion, SQL Queries & Database Operations

Deletion: Delete button per record with confirmation prompt. Example: DELETE FROM Guests
WHERE GuestID = 10;

SQL Operations Used: INSERT, SELECT, UPDATE, DELETE | WHERE, ORDER BY, GROUP BY |
COUNT(), SUM()

JOIN Example:

SELECT r.ReservationID, g.FirstName, g.LastName, rm.RoomNumber FROM Reservations r JOIN
Guests g ON r.GuestID = g.GuestID JOIN Rooms rm ON r.RoomID = rm.RoomID;

Screenshots: Before Deletion | After Deletion

TESTING, CONCLUSION & FUTURE ENHANCEMENTS

01

Testing Results

Insert: Passed
View: Passed
Update: Passed
Delete: Passed
Search: Passed
SQL Queries: Passed
DB Connectivity: Passed
FK Validation: Passed

02

Conclusion

Successfully built a database-driven hotel management system. MySQL delivers reliable CRUD operations. Relationships maintain data consistency and reduce manual effort.

03

Future Enhancements

Online booking & payment gateway. Guest login portal.
Email/SMS notifications.
Mobile app development.

04

Closing

Automated invoice generation. Advanced analytics and reports.
Thank You!
Questions?



Thank You!

QUESTIONS & DISCUSSION

PROJECT
Hotel Reservation & Guest
Service Management System

PRESENTATION
Database Management Systems –
Final Presentation | AY 2026–2027